Cloud Architecture Strategy

- Introduction
- Cloud-First
- Purpose-Built Platforms
  - Digital Edge
  - Cloud Data Store
  - Gatekeepers
  - Roles-as-a-Service (RaaS)
- Platform Brokerage
  - Multi-Cloud
- Management Control Planes and FinOps
  - FinOps
  - Service Catalog
  - Well-Architected Assessments
  - Multi-Tenancy and Vending Machines
- Integration-at-the-Edge
  - Joint Ventures
- Other items to consider....
  - Global Region Expansion
  - Security Assurance Framework
    - Cloud Security Posture Management Tools

Introduction

Cigna's new, evolved strategy is to be ***champions*** for ***affordable***, ***predictable***, and ***simple*** health care.

Aligning to the business strategy, one of the main themes of the IT strategy is ***modern and resilient technology***. One core mechanism to achieving modern and resilient solutions is to have a strong preference for **cloud-first architecture.**

Cloud-First

We should prefer cloud solutions as they offer benefits for resiliency, scale, and breadth-of-services.

Along with cloud-first architecture, other themes include:

- **Purpose-Built Platforms**
- **Platform Brokering**
- **Management Control Planes and FinOps**
- **Integration-at-the-Edge**

Purpose-Built Platforms

Cigna has created meta-platforms using multiple cloud services to **simplify** using cloud, with resiliency and security built into the DNA, which allows agile teams to deliver faster with the right guardrails "engineered in".

Digital Edge

A platform for managing Web API across distributed teams in a way that simplifies deployment and manageability. Details here

Cloud Data Store

A centralized Cloud Data Lake, with accompanied data moment patterns and tools, and a distribution model for other cloud accounts.

Gatekeepers

Gatekeepers are a way of extending deployment pipelines to the cloud, which allow for touchless deployments, using cost efficient processing in containers.

Roles-as-a-Service (RaaS)

RaaS is a way of managing least-privilege roles and policies at scale, using a GitOps-approach for defining, approving, and pushing custom roles and policy.

Platform Brokerage

Help teams make the best decision on which services to use from which world class providers

- Purpose-Built platforms
- SaaS
- Serverless
- Fully Managed Platforms / PaaS
- Cloud primitives: storage, network, compute
- Cloud Management Tools

Multi-Cloud

- Support for at least 2 first-tier public clouds (AWS / Azure)
    - Steerage based on functionality (Advanced AI, etc) and regulatory requirements, as well as workload fit
- Specialty Clouds as required by Regulation or Exception
    - Oracle Cloud: Helix and International
    - Alibaba: China business

Management Control Planes and FinOps

Provide modern, world-class tools to manage our solutions.

FinOps

- Tools to manage cloud spend and usage
- Spend anomaly detection
- Optimization opportunities

Service Catalog

- Chose from a library of optimized components

Well-Architected Assessments

- Evaluate solutions against a battery of tests
    - Reliability
    - Operational Excellence
    - Performance Efficiency
    - Cost Optimization
    - Security

Multi-Tenancy and Vending Machines

- Ability to quickly and easily isolate workloads to reduce "blast radius" and meet regulatory requirements
- Ability to still achieve economies of scale through automation and virtualization
    - account vending machines
    - namespace vending machine
    - cluster vending machine

Integration-at-the-Edge

Help teams move solve unique integration challenges like cloud-to-cloud, onprem-to-CSP, SaaS-to-SaaS quickly and efficiently, while protecting data. This may require new tools, patterns, and technologies

- Novel patterns with cloud-first components
    - For example, AWS PrivateLink support with SaaS and vendors
- Need to articulate more of the foundational benefits for this strategy iteration
- How do we handle JV integration in the cloud?

Joint Ventures

add detail

Other items to consider....

Global Region Expansion

- EU and APAC
- 2nd failover site in the US (us-east-2)

Security Assurance Framework

Fill in details from Mike McKenna's deck

Cloud Security Posture Management Tools

- Map compliance to frameworks like NIST, HIPAA, CIS, FedRAMP in near real-time
- Controls that manage the full lifecycle of solution
    - Directive: Establish governance, risk and compliance models.
    - Preventive: Protect workloads and mitigate threats/vulnerabilities
    - Detective: Provide full visibility and transparency over deployment operations.
    - Responsive: controls that drive remediation of potential deviations from your security baselines.